

Tsung-Yeh Hsieh

PhD Researcher | Scientific Machine Learning | AI for Physical Systems

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PROFILE/RESEARCH SUMMARY

CMU Mechanical Engineering PhD researcher developing scientific ML, probabilistic generative models, and GPU-accelerated simulation systems for physical and biological PDEs, with first-author work spanning surrogate modeling and physics-informed learning.

EDUCATION

Ph.D. in Mechanical Engineering, Carnegie Mellon University (Advisor: Prof. Jessica Zhang) Expected May 2029

M.S. in Power Mechanical Engineering, National Tsing Hua University 2023

B.S. in Power Mechanical Engineering, National Tsing Hua University 2021

SELECTED RESEARCH EXPERIENCE

Carnegie Mellon University - Graduate Researcher 2024 – Present

- Designed MGRD, a morphology-conditioned residual diffusion model for probabilistic spatiotemporal forecasting, reducing experimental microscopy MAE by 39.6% and improving skeleton F1 by 45.3% over a MetaFormer baseline with $\sim 99\times$ fewer parameters.
- Developed [GALDS](#), a graph-autoencoder + Neural ODE surrogate for 3D neurite transport, with $\sim 3\%$ mean and $< 8\%$ max relative error, while requiring $20\times$ less training data and $\sim 10\times$ fewer parameters and delivering $10\times$ faster inference than the PGNN baseline.
- Built [TubularFlowIGA](#), a C++/CUDA/MPI/PETSc pipeline from neuronal centerlines to hexahedral IGA meshes and GPU flow/transport solvers, achieving $36\text{--}154\times$ end-to-end speedups with CPU/GPU relative L_2 differences of approximately 10^{-6} .

National Tsing Hua University – Research Assistant 2021 – 2024

- Developed a stabilized physics-informed neural network with weak boundary-condition enforcement and transfer learning for advection-dominated PDEs; published as first author in *Engineering with Computers*.
- Developed POD-based reduced-order and differentiable simulation workflows for neural-network-controlled PDE systems, reducing simulation cost by up to 99%.
- Implemented GPU/CPU-parallel meshfree simulation workflows, achieving up to $4\times$ acceleration.

TECHNICAL SKILLS

Machine Learning: PyTorch, JAX, diffusion models, graph neural networks, Neural ODEs, transformers, autoencoders, physics-informed neural networks, surrogate and reduced-order modeling.

Scientific Computing: C/C++, CUDA, MPI, PETSc, FEniCS, isogeometric and finite element analysis, numerical PDEs, multiphysics simulation.

Software and HPC: NumPy/SciPy, MATLAB, Linux, Slurm, Docker, Git, ParaView.

SELECTED PUBLICATIONS

- [1] [TY. Hsieh](#), YJ. Zhang. "GALDS: A Graph-Autoencoder-based Latent Dynamics Surrogate model to predict neurite material transport". *Computer Methods in Applied Mechanics and Engineering*. (2025)
- [2] [TY. Hsieh](#), TH. Huang. "A multiscale stabilized physics informed neural networks with weakly imposed boundary conditions transfer learning method for modeling advection dominated flow." *Engineering with Computers* (2024)

HONORS

USNCCM Travel Award (2025) | NCFD Student Paper Competition, 3rd Place (2023)